

Mark Boss

Research Lead | Computer Vision & Graphics

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Mark Boss is the Co-Head of 3D & Image at Stability AI. He worked at Unity Technologies before and completed his PhD at the University of Tübingen in the computer graphics group of Prof. Hendrik Lensch. His research interests lie at the intersection of machine learning and computer graphics, focusing mainly on inferring physical properties (shape, material, illumination) from images.

Work Experience

Co-Head of 3D & Image - Stability AI

Oct. 2025 - present

Research Scientist - Stability AI

Jan. 2024 - Oct. 2025

Senior Research Scientist - Unity

Sep. 2022 - Jan. 2024

Ph.D. Student - University of Tübingen

Jun. 2018 - Jun. 2022

Student Researcher - Google

Jun. 2021 - Apr. 2022

Research Intern - NVIDIA

Apr. 2019 - Jul. 2019

Android Developer - zahlz

Jul. 2015 - Jun. 2017

Education

PhD in Computer Science

2023

University of Tübingen

MSc in Computer Science

2018

University of Tübingen

BSc in Computer Science

2016

Osnabrück University of Applied Sciences

Selected Publications

ReLi3D: Relightable Multi-view 3D Reconstruction with Disentangled Illumination

Jan. 2026 - In International Conference on Learning Representations (ICLR) - Jan-Niklas Dihlmann, Mark Boss, Simon Donne, Andreas Engelhardt, Hendrik P. A. Lensch, Varun Jampani

Stable Virtual Camera: Generative View Synthesis with Diffusion Models

Mar. 2025 - In ArXiv - Jensen Zhou, Hang Gao, Vikram Voleti, Aaryaman Vasishta, Chun-Han Yao, Mark Boss, Philip Torr, Christian Rupprecht, Varun Jampani

SF3D: Stable Fast 3D Mesh Reconstruction with UV-unwrapping and Illumination Disentanglement

Aug. 2024 - In Conference on Computer Vision and Pattern Recognition (CVPR) - Mark Boss, Zixuan Huang, Aaryaman Vasishta, Varun Jampani

SV3D: Novel Multi-view Synthesis and 3D Generation from a Single Image using Latent Video Diffusion

Mar. 2024 - In European Conference on Computer Vision - Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitrii Tochilkin, Christian Laforce, Robin Rombach, Varun Jampani

SAMURAI: Shape And Material from Unconstrained Real-world Arbitrary Image collections

Sep. 2022 - In Neural Information Processing Systems - Mark Boss, Andreas Engelhardt, Abhishek Kar, Yuanzhen Li, Deqing Sun, Jonathan T. Barron, Hendrik P. A. Lensch, Varun Jampani

NeRD: Neural Reflectance Decomposition from Image Collections

Nov. 2020 - In IEEE International Conference on Computer Vision - Mark Boss, Raphael Braun, Varun Jampani, Jonathan T. Barron, Ce Liu, Hendrik P. A. Lensch

